Ahmed Abdelkawy

Computer Vision&Machine Learning Researcher

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Research Interests

Computer Vision and Machine Learning. Specifically, action recognition, intelligent video surveillance, sports video analytics, and student engagement measurement.

Education

May 2022 - Current **Ph.D. in Electrical and Computer Engineering**, *University of Louisville*, Kentucky, USA.
GPA:3.9/4.0 | Expected Graduation: Dec 2026.

Oct 2019- May 2020 **PhD coursework in Computer Science**, *Helwan University, Faculty of Computers and Artificial Intelligence*, Cairo, Egypt.
GPA: 3.25/4.0

Oct 2013 - Oct 2018 **M.Sc in Computer Science**, *Helwan University, Faculty of Computers and Artificial Intelligence*, Cairo, Egypt.
Thesis: Real–time Abnormal Behavior Detection in Ubiquitous Environment

Sept 2008 - May 2012 **B.Sc in Computer Science**, *Helwan University, Faculty of Computers and Artificial Intelligence*, Cairo, Egypt.
Degree Grade: Excellent with honor – Class Rank: 1st | GPA: 3.8/4.0.

Publications

2026 **Ahmed Abdelkawy,** Ahmed Elsayed, Asem Ali, Aly Farag, Thomas Tretter, and Michael McIntyre. Context Matters: Peer-Aware Student Behavioral Engagement Measurement via VLM Action Parsing and LLM Sequence Classification. CVPRW 2026 (Accepted).

2025 **Ahmed Abdelkawy,** Asem Ali, and Aly Farag. EPAM-Net: An Efficient Pose-driven Attention-guided Multimodal Network for Video Action Recognition. Neurocomputing, 633:129781, 2025.

2024 **Ahmed Abdelkawy,** Aly Farag, Islam Alkabbany, Asem Ali,Chris Foreman, Thomas Tretter, and Nicholas Hindy. Measuring Student Behavioral Engagement using Histogram of Actions. Pattern Recognition Letters, 186:337–344, 2024.

2021 **Ahmed Nady** and Elsayed E Hemayed. Player Identification in Different Sport. In International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP) (5: VISAPP)(pp. 653-660), 2021.

2018 **Ahmed Nady,** Ayman Atia, and A Abutabl. Real-Time Abnormal Event Detection in Crowded Scenes. Journal of Theoretical and Applied Information Technology, 96(18):6064–6074,2018.

Research & Industry Experience

May 2022 – Current **Graduate Research/Teaching Assistant**, *CVIP Lab, University of Louisville.*

- Developed and implemented novel deep learning architectures (e.g., EPAM-Net) using PyTorch for computationally efficient multimodal human action recognition, addressing limitations for real-time deployment and achieving up to 72.8x reduction in FLOPs.

- Developed a novel framework using a 'histogram of actions' derived from student's actions and gaze to quantify student behavioral engagement.
- Designed and curated a new benchmark dataset for student behavioral engagement, including video annotation for 13 actions and 4 learning-centered emotions over 1400 video segments and 1200 images, respectively.
- Assist in teaching courses including Artificial Intelligence Systems, Pattern Recognition & Machine Intelligence, and Digital Signal Processing. Duties include grading coursework and providing selected lectures.

Sept 2018 - Mar 2020 **Computer Vision Engineer (Part-time)**, *SawaSports Analytics startup company*, Cairo, Egypt.

(Demo:`/www.youtube.com/watch?v=Jwyq9B87tS8`).

- Developed modules for player/ball detection and tracking using YOLO object detector and DeepSORT.
- Designed and implemented a Convolutional Neural Network (CNN) architecture for player identification based on jersey number.
- Modeled and implemented recognition of key game events (passes, corners, shots on goal) using a Finite State Machine (FSM) approach.
- Implemented a player skill assessment module using pose estimation (skeleton tracking) and biomechanical phase analysis from video.

Oct 2016 - Aug 2018 **Graduate Researcher (MSc Thesis)**, *Helwan University, Faculty of Computers and Artificial Intelligence*, Cairo, Egypt.

- Investigated and developed unsupervised machine learning techniques for abnormal event detection in crowd videos, achieving real-time performance (25 fps) and high accuracy (98.48% AUC on UMN dataset).

Jan 2016 - Aug 2016 **Software Engineer (Part-time)**, *Open Minds CIT*, Cairo, Egypt.

- Maintained and improved the performance of the Vehicle Access Control System.
- Designed and developed a Traffic Monitoring System by utilizing automatic number plate recognition.

Sept 2014 - Apr 2015 **Image Processing Software Engineer (Part-time)**, *Farminal company*, Cairo, Egypt.

- Developed a thermal image-based technique for early detection of mastitis (udder inflammation) in dairy cow using Udder Region segmentation and time series analysis.

Technical Skills

Programming Langauages: Python, Java, C/C++, Matlab, CUDA C

Libraries/ Framework: Pytorch, Tensorflow, Keras, OpenCV, OpenGL

Academic Service

Reviewer.

- International Journal of Computer Vision (IJCV).
- Neurocomputing.
- Pattern Recognition.
- Pattern Recognition Letters.

Teaching Experience

Dec 2013 - April 2022 **Helwan University, Faculty of Computers and Artificial Intelligence**, *Cairo*, Egypt.

Assistant Lecturer (Dec. 2018 – April 2022)
Teaching Assistant (Dec. 2013 – Nov. 2018)

Assisted in teaching courses such as Introduction to Artificial Intelligence, Programming Language, Natural Language Processing, Data Structure, Introduction to Algorithms, and Pattern Recognition, responsibilities included grading, leading lab sessions.